Iridescent Lesson Plan

PRE-PLANNING

OBJECTIVE What will your students be able to do?	KEY CONCEPTS AND VOCABULARY What three-five key points will you emphasize?
To understand the high-level concepts behind the working principles of cellular networks and sensor networks.	• Sensors, Networks • Data communications

CONNECTION TO THE BIG IDEA
How does the objective connect to the big idea?

Learning about different kinds of sensors and networks will make them appreciate the so many interesting applications that are possible. As a proof of concept, we will make a temperature monitoring application.

ASSESSMENT
How will you know whether your students have made progress toward the objective? How and when will you assess mastery?

Exit slips and concept maps will enable us to check for student understanding.

LESSON

OPENING (2 min)
How will you communicate *what* is about to happen? How will you communicate *how* it will happen? How will you communicate its *importance*? How will you communicate *connections* to previous lessons?

Good morning everyone! Let us review what we have learnt so far. Do you remember the two primary components that are fundamental to wireless communications. Right! Transmitters and Receivers. And in the last two lessons, we built primitive versions of them. Do you remember what they were? (i) a spark-gap transmitter, which we used to produce radio waves, and (ii) a crystal radio receiver, which we used to build a foxhole radio. Now, do you think there are other components to a wireless system that make this communication between a transmitter and receiver possible? We saw some pictures of these components earlier. Yes, they are called "Radio Towers" or "Base Stations", which do a lot of work by "sitting" in between the transmitters and receivers in order to send messages to the "right" places.

(Show slide 1 to give an example of how cellular phones work at a very high level to motivate. Talk about the complexity it takes to correctly route tens or even hundreds of calls made at the same time from a given location. For example, if A is calling to B, and C is calling to D, and both these calls are made from the same house, how the base station figures out that it has to "connect" A to B and not to C. Elaborate.)

Today we will learn about the concept of a "network", and how wireless communications are possible without involving these big radio towers. We will also learn about "sensors", and how we can combine them with transmitters and receivers in order to make some interesting applications that we can use in our daily lives. Can you guess what we might call such a network that involves sensors? Right! "Sensor Networks"!

MATERIALS
What materials will you need for your lesson?

(for each experiment)
- Pre-programmed sensor nodes, one for each student
- Laptop
- Candles, small ice bags, small zip-lock plastic bags

DIRECT INSTRUCTION (8 min)

What key points will you emphasize and reiterate?
How will you ensure that students actively take-in information?
How will you vary your approach to make information accessible to all students?
Which potential misunderstandings will you anticipate?
How will your students be using a Concept Map or other structured tool?

First talk about the concept of a "network". Ask them if they have heard about computer networks, such as the Internet (which is the biggest global network of computers connected to each other). More specifically, we can define "network" a set of electronic devices (computers, cell phones, etc), which are "interconnected" and perform some functions for its users. Give examples of networks by showing the following slides.

Project Slide 2 that shows a cell phone network. Point out the way the base stations (BTS: stands for Base Transceiver Station) are connected with each other, and through some boxes labeled as BSC (stands for Base Station Controller, we don't have to tell them the full form unless they ask for it) and MSC (Mobile Station Controller, again we don't have to tell them the full form) to the Internet. Also, point out how two pairs of cell phones (A to B and C to D) talk to each other by first connecting to their "closest" base stations. Ask the students if they can give a definition of a "cell phone network" just by looking at the picture. Now, ask them if they think it might be possible to remove the base stations (the towers) and the boxes and still be able to communicate.

Project Slide 3 that shows a sensor network, and ask them the difference between this and the previous network (just from visual appearance). They should at least be able to point out that the base stations are gone, and the small circles (called sensor nodes) are connected "directly" to each other as opposed to in a cellular network.

Project Slide 4 that shows a sensor node (each of those small circles from the previous slide). Also, bring such a sensor node to class and pass it on so that everyone can see it closely how it looks in reality.

Project Slide 5 that shows more sensor nodes (Tmote Sky, Smart Dust, Sun SPOT, MICA2); bring these to class too if possible. Tell them that these tiny devices carry different kinds of sensors, which can measure temperature, light, humidity, motion, sound, etc. Ask them if they can think of any real-life applications where they could build a network using these sensors.

Project Slides 6, 7, 8 that show a few motivating applications (agricultural crop monitoring, talking plants, bridge monitoring, wild-fire monitoring, earthquake early warning). Ask them how these applications might operate. Explain at a very high level how wireless communication is used in these applications to transmit measurements.

DIRECTIONS FOR EXPERIMENT(5 min)

How will you clearly state and model behavioral expectations?
How will you give 3-5 clear directions for the activity and model them?

We are going to show the students how to form a network using a bunch of temperature sensors, and how they can communicate with each other and exchange temperature measurements without the help of any base station. As a second goal, we will also demonstrate that two sensor nodes can communicate and exchange data only if they are within the transmission range of each other.

EXPERIMENT (40-60 min)

What kind of activity can be performed by students that directly relates to your objective and the big idea?
How will you engage students and capture their interest to explore the concept?
What exactly will your students be doing during the activity? What will you be doing?
How will your students be using a Concept Map or other structured tool?

The demonstration will run as follows:

1. *We will distribute each student a pre-programmed sensor node along with two AA batteries. Initially, the batteries are not yet connected to the nodes. The demonstrator will attach one node to the USB port of a laptop. This node, usually called a "sink", runs using the laptop's power source (i.e., it does not require a battery), and will be used to collect data from all other nodes. We will ask the students to sit close to the laptop to make sure that all the nodes are within the transmission range of the sink.*

2. *Each node is identified by a number (an integer), which is stored in the node's memory. We will tell each student what numbers their nodes contain while distributing the nodes in Step 1. Each node can also measure the temperature of its surroundings. The number stored in the sink is 0.*

3. *We will divide the students into three equal groups: A, B, and C, and distribute: (i) small ice-bags to every student in group A, and (ii) candles & match boxes to every student in group B. Students in group C just have the sensor nodes. We will also give small plastic bags (zip lock bags are ideal) to all those who got ice bags.*

4. *Initially all nodes are switched off (because the batteries are not yet connected). We will now ask them to connect the batteries and quickly put the nodes inside the small bags and then inside the ice bags, or hold them near the candle flames (at least 5 inches away from the flame to make sure that they don't burn the nodes). Students in group C just put the batteries on. The idea is to keep a sensor node close to a heat source, a cold source, or just normal so that they measure different temperatures.*

5. *Once a node is turned on by connecting the battery, it will measure the temperature around itself and will start transmitting packets containing the temperature readings. All nodes that are within the transmission range of each other will receive packets, but only the sink node will record the temperature.*

6. *Now we will ask two students from each group to move away from the laptop holding the nodes along with the candles and ice bags in their hands. The idea is that as the nodes move away from the laptop, some of their transmitted packets are not received by the sink (they just get lost), and as they continue to move further away none of the packets will be received.*

7. *Ask everyone to disconnect the batteries. The demonstrator will now read off data from the sink node that is connected to the laptop. The sink is programmed to store the number of packets (containing temperature readings) and the node identifier for*

each node. We will show that the sink has received some low temperatures, some high temperatures, and some room temperatures depending on the heat or ice source. We will also show that the number of packets received from nodes that were moving away from the laptop is much smaller than others which were stationary.

REFLECTION (5 min)

How will students summarize what they learned?
How will students be asked to state the significance of what they learned?
How will students relate what they learned back to your objective and big idea using key vocabulary?

Exit Slip questions:

1. If you put the nodes inside your bags, will the packets transmitted by them be still received by the sink node connected to the laptop?
2. Will the packets still be received if you were in another room separated by a wall but at same distance from the laptop (for example, suppose that the laptop is placed next to the wall in one room, and you are in the adjacent room close to the wall)?
3. What can we do to make sure that the nodes moving away from the laptop are still able to send packets successfully to the sink node connected to the laptop? Do you think we should re-program the sensor nodes that are moving away or the sink node? What are we going to change while re-programming?
4. What is the most important thing you learnt today?